



Cyber-Physical Systems
in Mechanical Engineering TU Berlin

Project Introduction

TU Berlin Summer School 2026 – DL4TS

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Group Projects



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Idea:

- Form groups of 2 participants
- Choose one time-series modeling project for the course
- Use the practical sessions to implement and evaluate your models

Workflow:

- Implement a baseline model
- Add an advanced method
- Compare with a foundation model

Goal:

- Compare model performance and effort
- Discuss strengths and limitations of each approach
- Present your findings to the course

Group Projects



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1. Sensor correction (Sequence-to-Sequence regression)
2. Electricity dataset forecasting (Long-Horizon Forecasting)
3. FordA or FordB engine state classification (Classification)
4. ECG5000 (Classification)
5. NASA C-MAPSS sequence-to-one regression (Predict remaining lifetime)
6. Duffing Oscillator (physics informed forecasting with exogenous input)
7. Your own projects

1. Data-Driven Sensor Signal Correction



Setting:

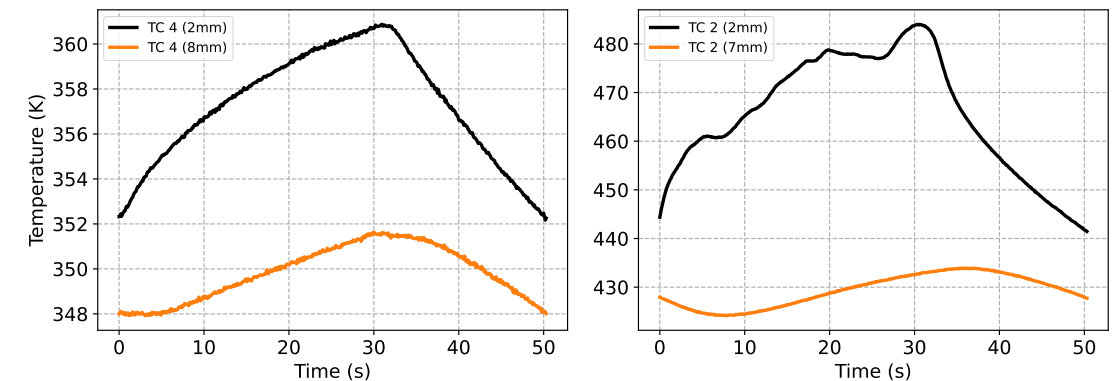
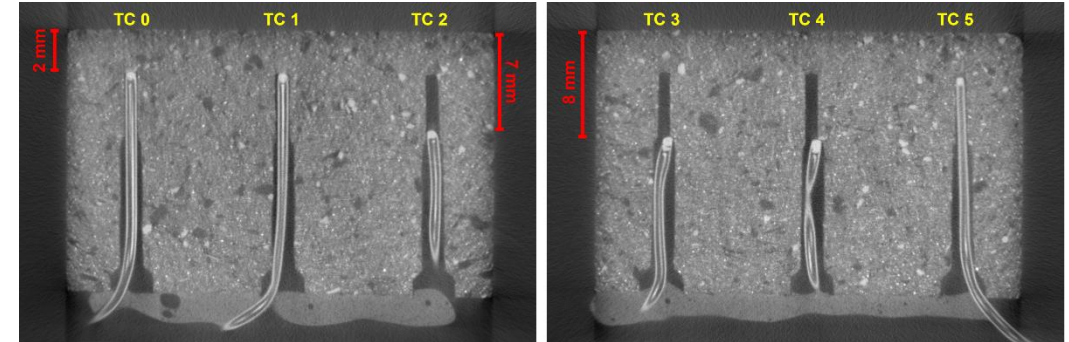
- Mispositioned thermocouple sensors
- Systematically lower temperature readings

Dataset:

- 211 samples with
- Multivariate time series (12 features)
- Variable length time series

Goal:

- Reconstruct the biased signals from the correctly installed sensors
- Map the 9 correctly installed sensors to the 3 mispositioned



2. Electricity Dataset Forecasting



Setting:

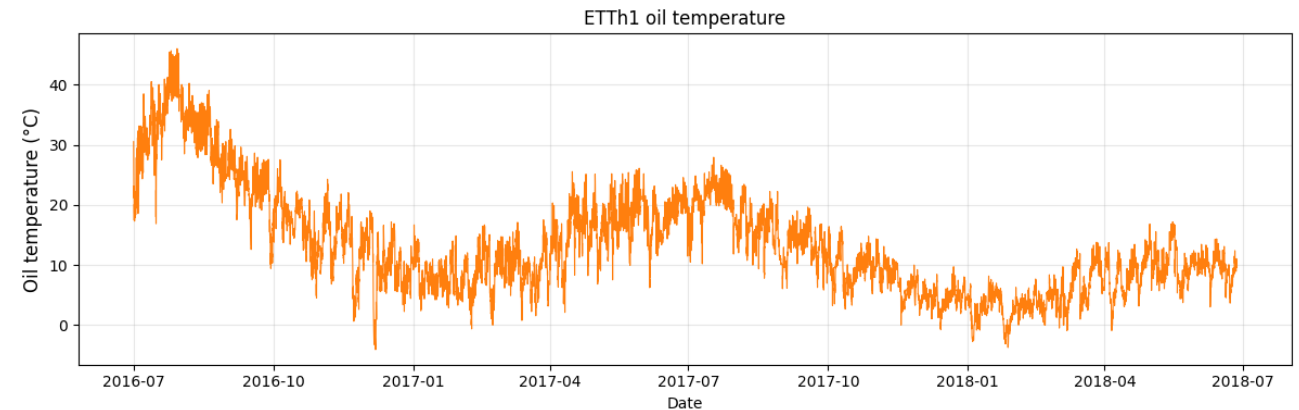
- Common benchmark for long-horizon time-series forecasting
- Transformer load and oil temperature measured

Dataset:

- 17,420 hourly time steps (2 years of data)
- 7 numerical time-series variables
- Additional data column

Goal:

- Forecast the oil temperature
- $\mathbf{X}_{t-L+1:t} \in \mathbb{R}^{L \times 7} \rightarrow \hat{\mathbf{y}}_{t+1:t+H} \in \mathbb{R}^H$



3. Motor Engine Symptom Detection



Setting:

- Competition in the IEEE World Congress on Computational Intelligence
- Motor engines symptoms

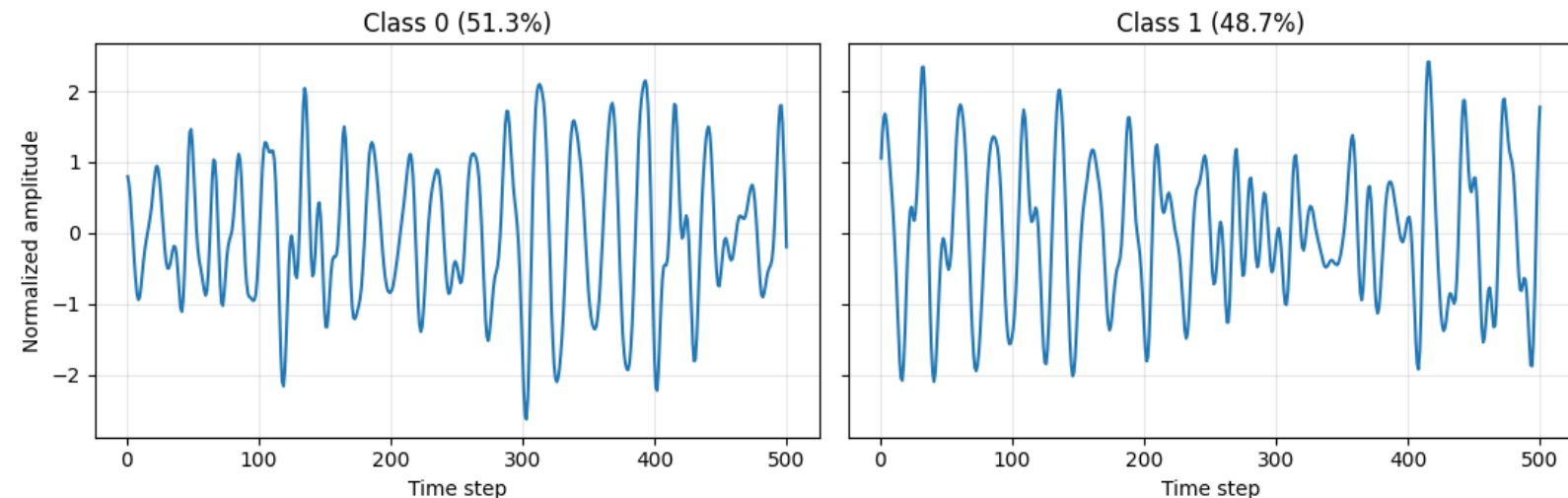
Dataset:

- 3601 train samples
- 1320 test samples
- Univariate time series
- FordA and FordB

Goal:

- Predict motor engine symptoms from noisy dataset
- FordB dataset with more noise

Example FordA signals



4. ECG Waveform Classification



Setting:

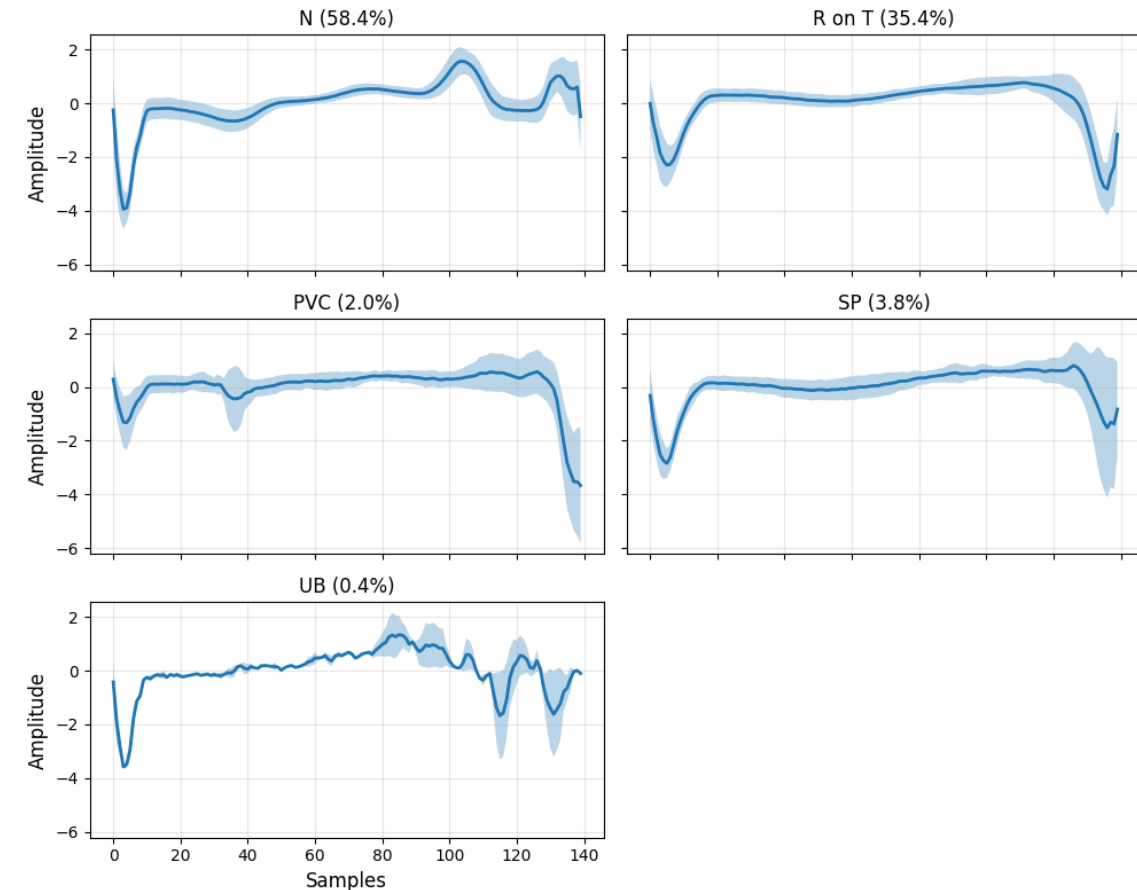
- Patient with severe congestive heart failure
- 20-hour long ECG

Dataset:

- 5 classes
- High class imbalance
- 500 train samples
- 4500 test samples

Goal:

- Classify the ECG waveform signals
- Deal with an imbalanced dataset



5. RUL Prediction for Turbofan Engines



Setting:

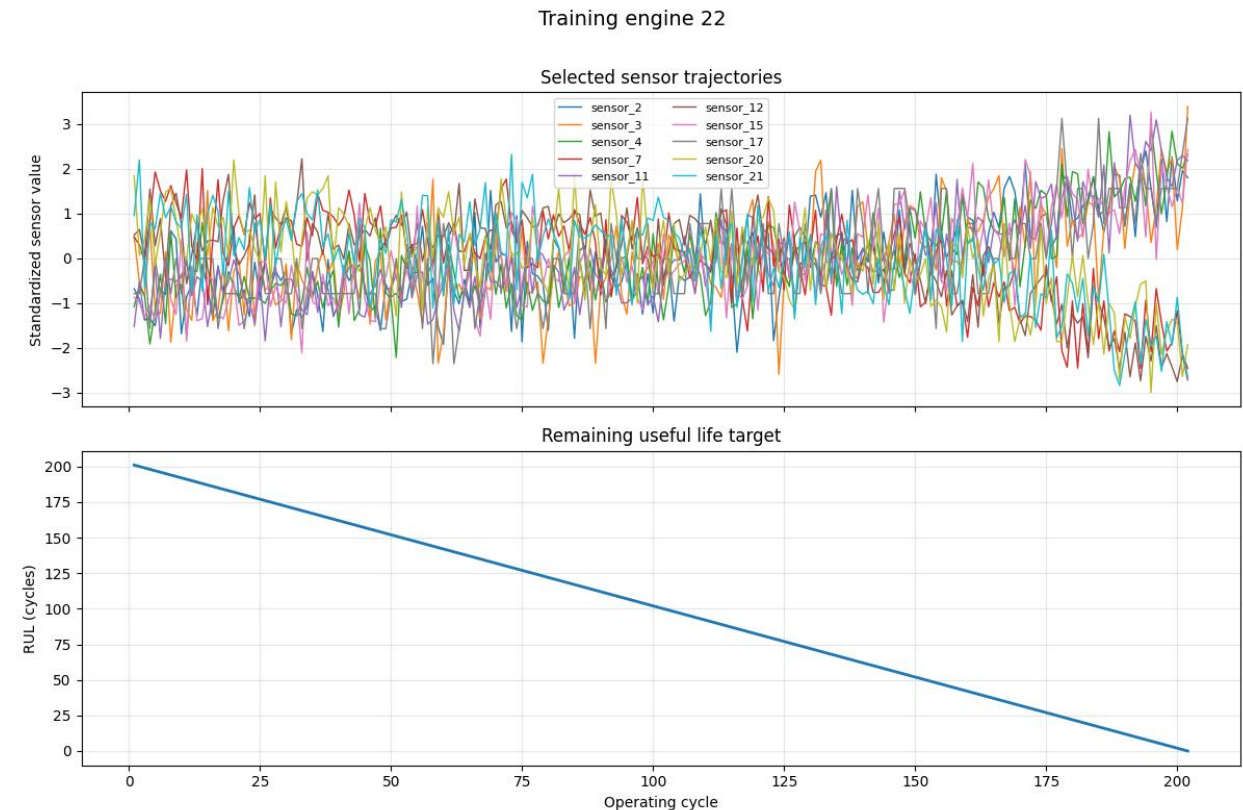
- Remaining Useful Lifetime (RUL) prediction
- Simulated data from NASA

Dataset:

- Data from 200 engines (50/50, train/test)
- 21 sensor readings
- 3 operating settings
- FD003 includes the RUL for the Fan
- Variable number of operating cycles

Goal:

- Predict remaining useful lifetime for engine (+Fan)



6. Duffing Oscillator Response Analysis



Setting:

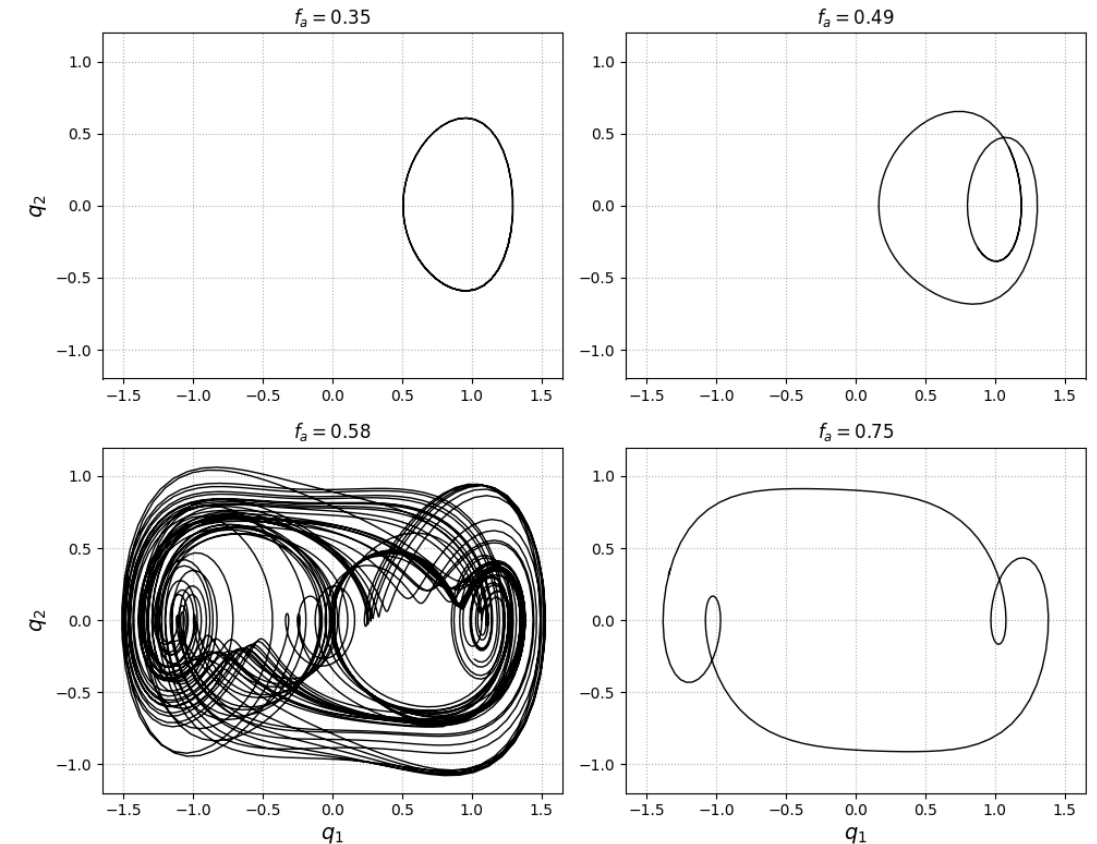
- Nonlinear oscillator to study complex dynamics
- Multiple stable states and bifurcations
- Chaotic motion

Dataset:

- 2 trajectories from the period-2 cycle regime
- 2500 time steps each
- External forcing

Goal:

- Predict period-1,2,4 and chaotic regime
- Coupling of a Physic-Informed Neural Network (PINN)



7. Your own projects



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If you have your own dataset that you want to work on, we are happy to support you with it!